

TIANSHUI, China

WMO: 570060

Lat: 34.577N

Lon: 105.752E

Elev: 1143

StdP: 88.33

Time Zone: 8.00 (E08)

Period: 90-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.1	-6.6	-17.4	0.9	-1.2	-15.1	1.2	-1.3	4.5	2.3	3.7	2.3	0.2	250	0.387

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	32.3	19.7	30.8	19.4	29.6	19.2	22.0	28.3	21.4	27.4	20.7	26.7	2.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.3	17.3	24.9	19.6	16.4	24.4	18.8	15.7	24.0	71.0	28.4	68.4	27.5	65.8	26.7	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.4	3.6	3.2	-11.2	35.0	2.2	1.8	-12.8	36.3	-14.1	37.3	-15.3	38.3	-16.9	39.6		
(5)			-12.1	23.4	2.1	0.8	-13.6	24.0	-14.8	24.4	-16.0	24.9	-17.5	25.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.2	-0.6	2.7	7.5	13.5	17.9	21.6	24.2	22.9	18.1	11.9	6.0	0.5		
	DBStd	9.04	2.43	2.98	3.57	3.91	3.24	2.41	2.43	2.63	3.29	3.14	3.35	2.64		
	HDD10.0	1078	329	204	93	12	1	0	0	0	0	18	127	295		
	HDD18.3	2729	587	438	334	152	46	3	0	2	43	198	371	553		
	CDD10.0	1893	0	0	17	116	245	348	441	399	244	78	6	0		
	CDD18.3	503	0	0	0	6	32	102	183	142	37	1	0	0		
	CDH23.3	3962	0	0	3	69	301	788	1539	1037	223	2	0	0		
	CDH26.7	1257	0	0	0	9	63	227	576	331	51	0	0	0		
(14) Wind	WSAvg	1.0	0.7	1.0	1.3	1.2	1.1	1.1	1.1	1.1	0.9	0.7	0.7	0.7		
(15) Precipitation	PrecAvg	539	4	6	18	44	57	67	95	88	92	47	15	4		
	PrecMax	773	11	17	42	81	121	174	195	218	180	114	41	18		
	PrecMin	325	0	0	0	11	11	14	26	16	34	13	3	0		
	PrecStd	122	3	4	9	21	29	36	41	53	45	25	10	4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.8	15.3	23.0	27.8	30.4	33.0	34.9	33.6	31.1	23.1	17.9	9.6		
		MCWB	3.1	6.5	11.5	15.7	16.1	18.7	20.7	20.1	17.8	13.7	10.5	3.7		
	2%	DB	7.4	12.1	19.0	25.2	28.4	30.7	32.7	31.5	28.2	20.6	15.7	8.0		
		MCWB	1.5	5.2	9.4	14.2	15.4	18.1	20.2	20.0	18.7	13.4	9.5	2.8		
	5%	DB	5.5	10.0	16.4	23.1	26.6	29.4	31.2	30.0	26.0	19.0	13.9	6.6		
		MCWB	0.6	3.7	8.5	12.8	15.0	18.0	20.1	20.0	18.1	12.9	8.5	1.8		
	10%	DB	3.9	8.1	14.0	20.8	24.8	27.9	29.8	28.5	24.0	17.1	11.8	5.1		
		MCWB	-0.3	2.7	7.2	11.7	14.6	17.7	20.0	19.7	17.3	12.2	7.6	0.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	7.1	13.4	16.9	18.5	20.8	23.2	22.8	21.7	15.7	11.7	5.0		
		MCDB	8.8	13.5	20.8	25.1	25.9	27.8	30.9	29.0	26.5	19.7	16.3	8.5		
	2%	WB	1.9	5.8	10.2	15.4	17.3	19.9	22.3	21.9	20.1	14.6	10.3	3.3		
		MCDB	6.5	11.3	17.7	23.0	24.8	26.8	29.0	28.0	25.8	18.9	14.3	6.8		
	5%	WB	0.8	4.4	8.9	13.9	16.3	19.3	21.6	21.3	18.7	13.5	9.3	2.2		
		MCDB	5.1	9.2	15.2	21.8	23.7	26.2	27.9	27.2	24.2	17.5	12.9	5.8		
	10%	WB	-0.2	3.2	7.8	12.6	15.5	18.8	21.0	20.6	17.7	12.6	7.9	1.2		
		MCDB	3.4	7.0	13.3	19.3	22.4	25.5	27.2	26.3	22.8	16.3	11.4	4.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.3	8.5	9.4	10.3	10.7	10.4	9.8	9.2	8.3	8.2	8.7	8.1		
		MCDBR	11.9	12.6	14.4	14.6	14.8	14.0	12.6	12.2	11.6	11.2	11.9	10.9		
	5% WB	MCWBR	7.3	7.0	6.8	6.1	5.1	4.5	3.5	3.6	4.1	5.1	6.6	6.8		
		MCWBR	10.9	11.7	12.9	12.6	12.1	10.9	9.7	9.8	9.9	9.8	10.9	9.3		
(40) Clear-Sky Solar Irradiance	taub		0.472	0.541	0.588	0.588	0.565	0.547	0.549	0.588	0.551	0.532	0.475	0.432		
	taud		1.790	1.677	1.626	1.664	1.784	1.882	1.935	1.810	1.853	1.821	1.858	1.904		
	Ebn at Noon		679	678	692	723	748	759	755	715	717	681	673	694		
	Edh at Noon		171	214	244	246	221	200	188	210	192	182	158	144		
(44) All-Sky Solar Radiation	RadAvg		2.59	3.08	4.09	5.04	5.33	5.31	5.06	4.56	3.32	2.88	2.58	2.43		
	RadStd		0.25	0.45	0.37	0.47	0.45	0.37	0.65	0.54	0.32	0.41	0.38	0.30		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page